

Lab-3 (2) InterVLAN Routing(E)

07112303

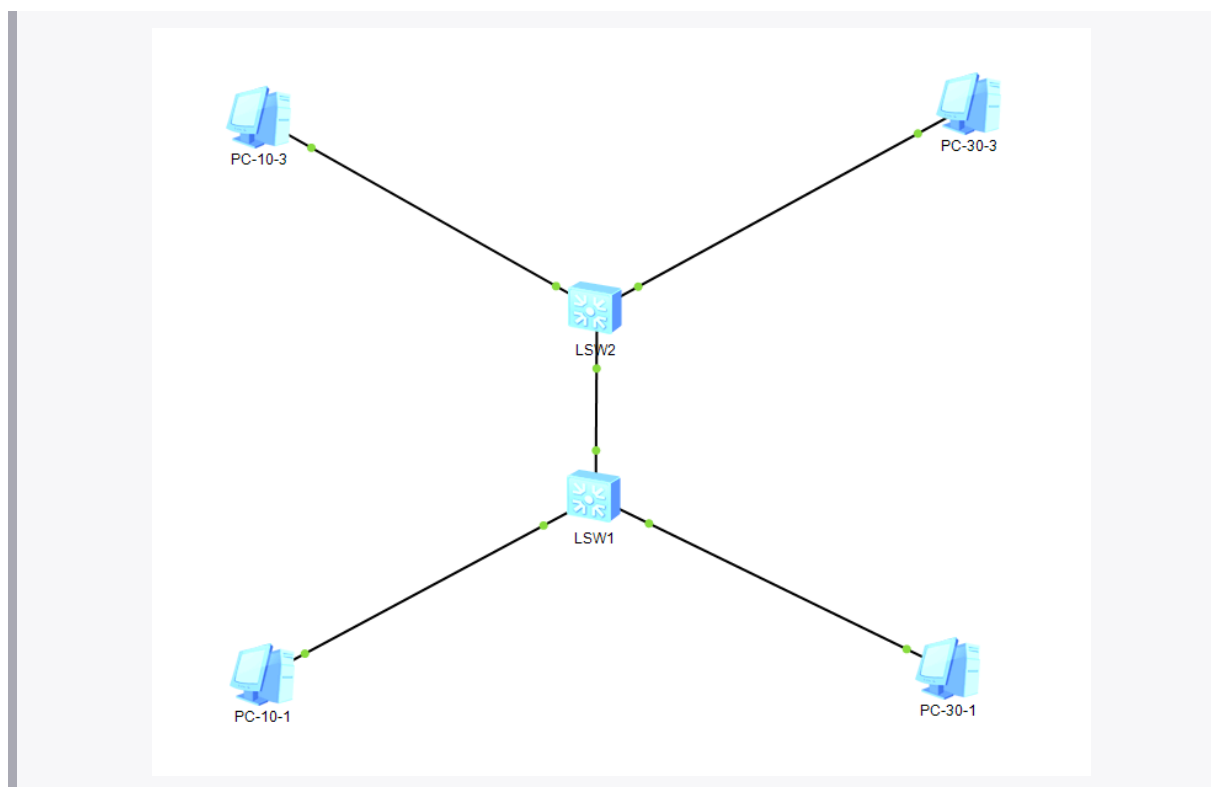
左逸龙

1120231863

2026-05-22

Lab-3-1 VLANIF CONFIGURATION

Q3-1.1. Paste the screenshot of the created topology.



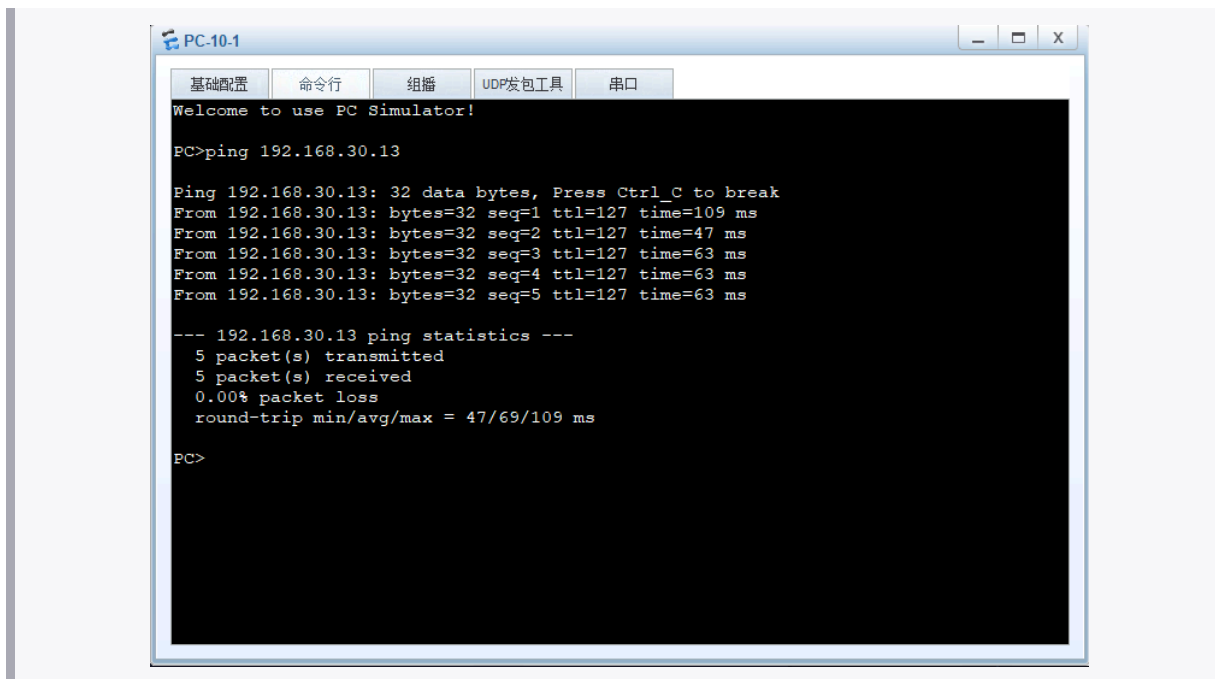
Q3-1.2. Paste the screenshot of the IP routing table on the switch LSW1.

```
[LSW1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 6          Routes : 6

Destination/Mask    Proto    Pre  Cost    Flags NextHop         Interface
-----
127.0.0.0/8         Direct   0     0        D  127.0.0.1       InLoopBack0
127.0.0.1/32        Direct   0     0        D  127.0.0.1       InLoopBack0
192.168.10.0/24      Direct   0     0        D  192.168.10.1    Vlanif10
192.168.10.1/32      Direct   0     0        D  127.0.0.1       Vlanif10
192.168.30.0/24      Direct   0     0        D  192.168.30.1    Vlanif30
192.168.30.1/32      Direct   0     0        D  127.0.0.1       Vlanif30
```

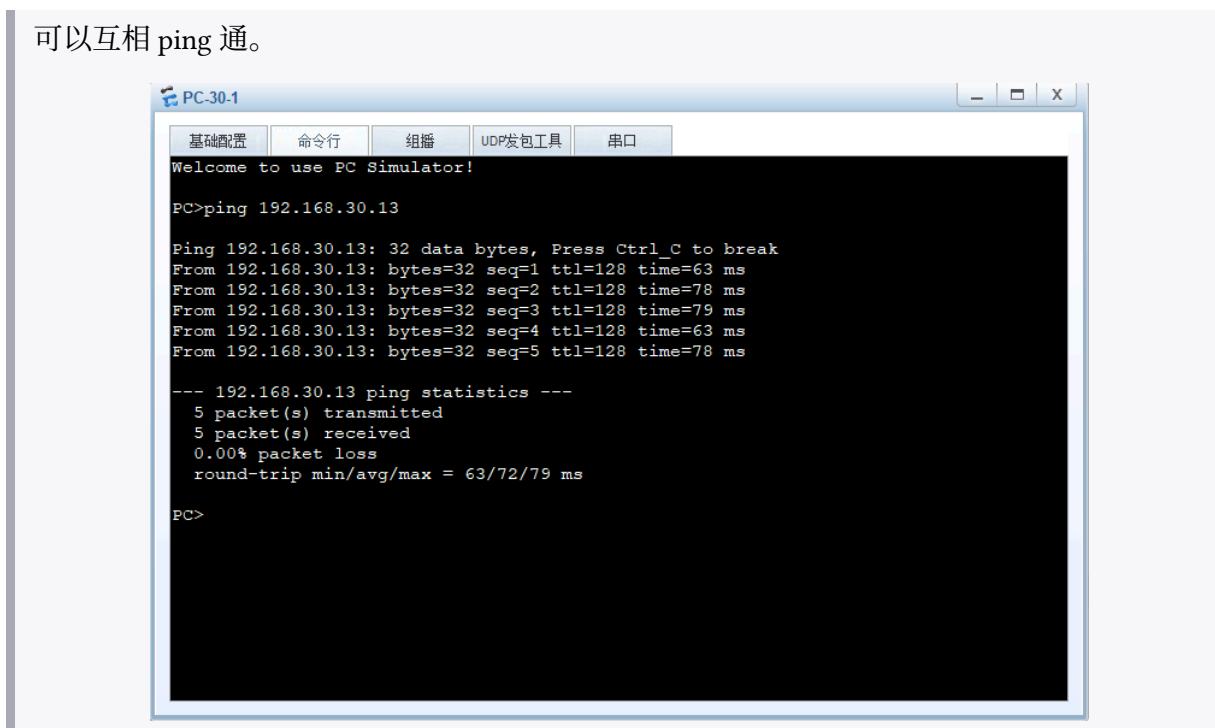
Q3-1.3. Can PC-10-1 and PC-10-3 ping each other? Please paste the screenshot of the ping result.

可以互相 ping 通。



Q3-1.4. Can PC-30-1 and PC-30-3 ping each other? Please paste the screenshot of the ping result.

可以互相 ping 通。



Q3-1.5. Can PC-10-1 and PC-30-3 ping each other? Please paste the screenshot of the ping result.

可以互相 ping 通。

```
PC>ping 192.168.30.13

Ping 192.168.30.13: 32 data bytes, Press Ctrl_C to break
Request timeout!
From 192.168.30.13: bytes=32 seq=2 ttl=127 time=78 ms
From 192.168.30.13: bytes=32 seq=3 ttl=127 time=78 ms
From 192.168.30.13: bytes=32 seq=4 ttl=127 time=62 ms
From 192.168.30.13: bytes=32 seq=5 ttl=127 time=78 ms

--- 192.168.30.13 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/74/78 ms
```

(注：由于跨网段通信时需要先通过 ARP 协议获取网关的 MAC 地址，因此第一个 ICMP 请求出现超时属于正常现象。)

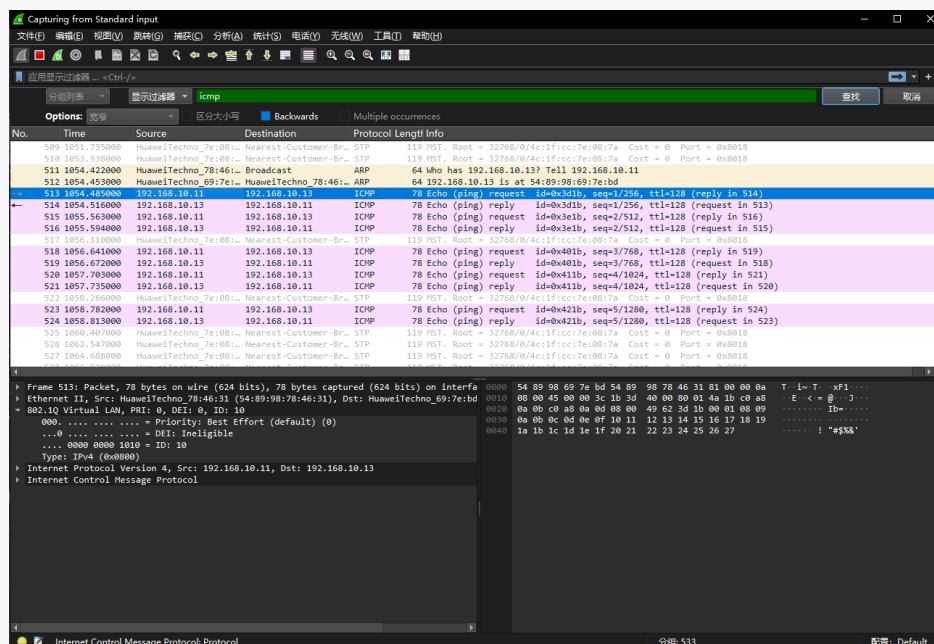
Q3-1.6. Please paste the screenshot of the result of command “tracert 192.168.30.13” issued from PC-10-1.

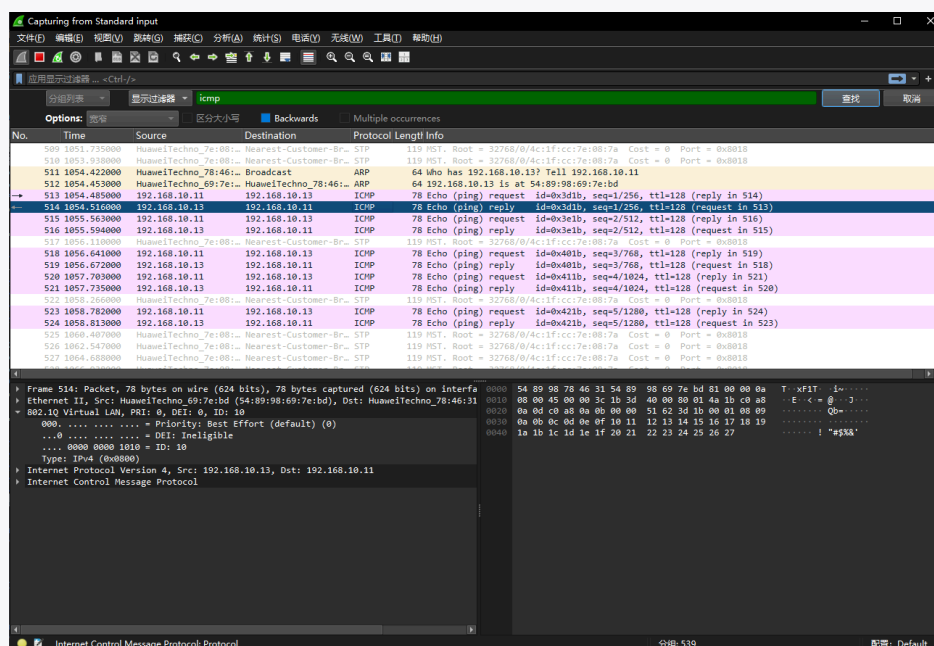
```
PC>tracert 192.168.30.13

tracert to 192.168.30.13, 8 hops max
(ICMP), press Ctrl+C to stop
 1 192.168.10.1 32 ms 15 ms 16 ms
 2 192.168.30.13 78 ms 63 ms 62 ms
```

Q3-1.7. Analyze the captured ping traffics between the PC-10-1 and PC-10-2. What are the tagged VLAN IDs of the Ethernet frames leaving and entering the port? Please paste the screenshot of the all field information of the captured Ethernet frame.

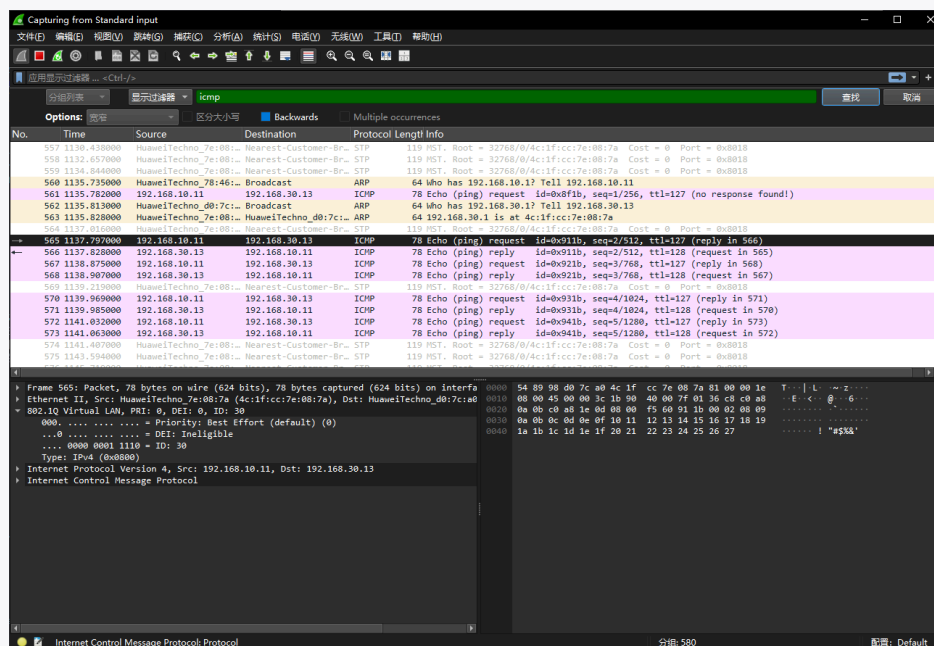
(注：按照实验实际保留的设备，此处使用 PC-10-1 与 PC-10-3 进行抓包分析。)

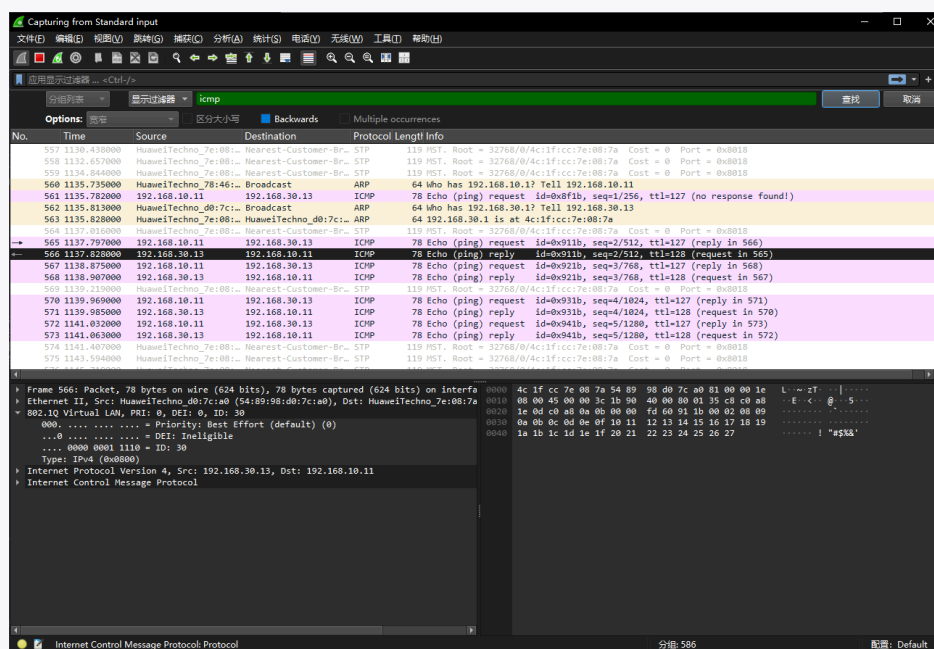




捕获到的 Ethernet 帧进入和离端口口的 VLAN ID 均为 10。因为 PC-10-1 和 PC-10-3 同属于 VLAN 10，它们之间的通信属于二层交换转发，不经过三层路由。因此，在两台交换机之间的 Trunk 链路上，数据帧仅携带源 VLAN（即 VLAN 10）的标签。

Q3-1.8. Analyze the captured ping traffic between the PC-10-1 and PC-30-2. What are the tagged VLAN IDs of the Ethernet frames leaving and entering the port? Please paste the screenshot of the all field information of the captured Ethernet frames.



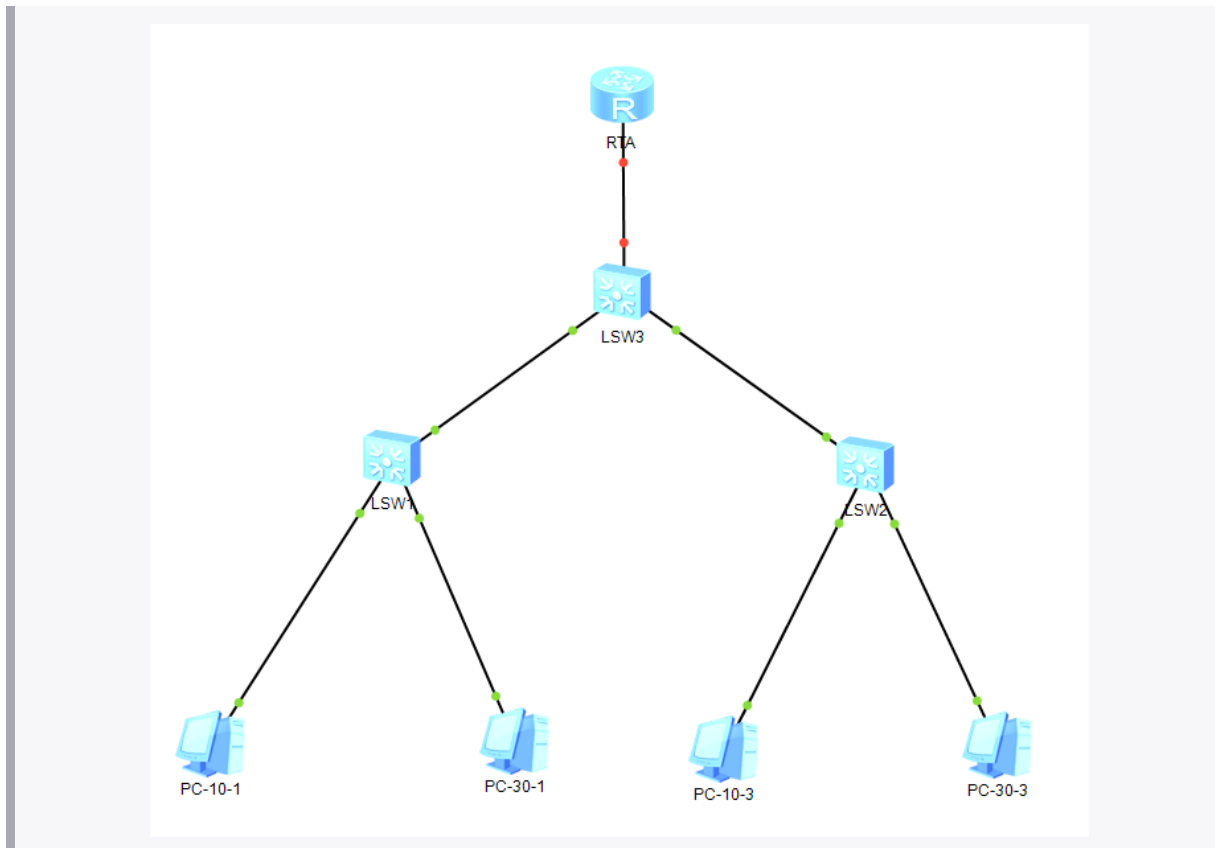


捕获到的 Ethernet 帧进入和离开 LSW1 的 Trunk 端口时的 VLAN ID 均为 30。原因如下：

- 数据报文从 PC-10-1 发出到达 LSW1 时，属于 VLAN 10。
- 因为目的 IP 属于另一个网段，LSW1 查找内部路由表进行三层转发，将报文从 VLAN 10 解封封装，并重新打上 VLAN 30 的标签。
- 随后，这个已经被打上 VLAN 30 标签的报文才从 LSW1 的 GE 0/0/24 接口离开并发送给 LSW2。因此我们在该接口抓到的跨 VLAN 数据包，其 802.1Q 标签已经被修改为了目的网络的 VLAN ID。

Lab-3-2 ONE-ARMED ROUTER CONFIGURATION

Q3-2.1. Paste the screenshot of the created topology.



Q3-2.2. Paste the screenshot of the IP routing table on the router RTA.

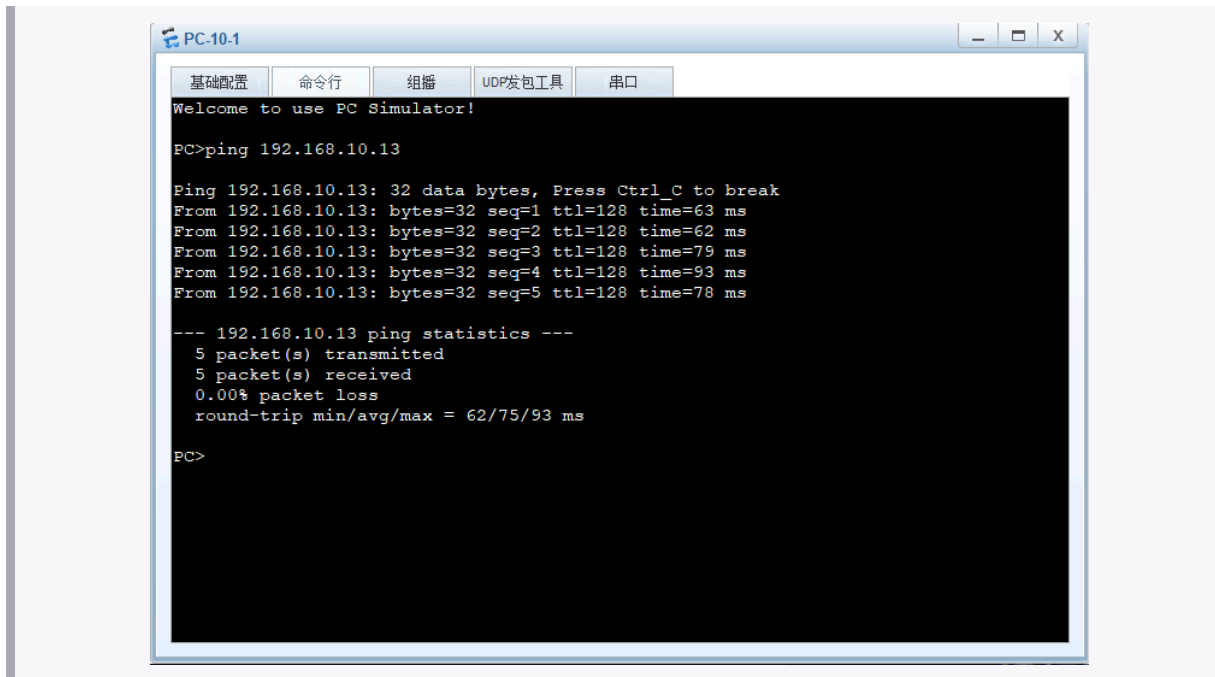
```

[RTA]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
    Destinations : 10          Routes : 10

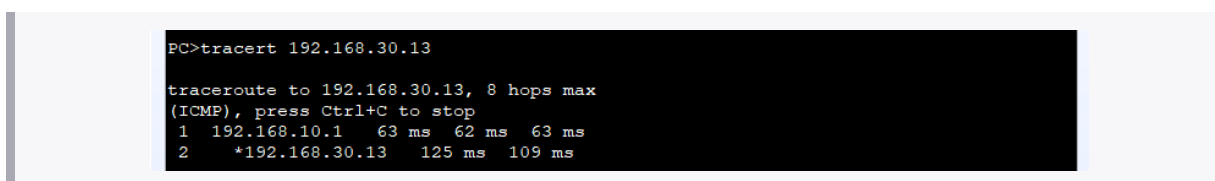
Destination/Mask    Proto    Pre  Cost    Flags NextHop         Interface
-----
    127.0.0.0/8      Direct  0     0        D    127.0.0.1         InLoopBack0
    127.0.0.1/32     Direct  0     0        D    127.0.0.1         InLoopBack0
127.255.255.255/32  Direct  0     0        D    127.0.0.1         InLoopBack0
    192.168.10.0/24   Direct  0     0        D    192.168.10.1       GigabitEthernet
0/0/2.10
    192.168.10.1/32   Direct  0     0        D    127.0.0.1         GigabitEthernet
0/0/2.10
    192.168.10.255/32 Direct  0     0        D    127.0.0.1         GigabitEthernet
0/0/2.10
    192.168.30.0/24   Direct  0     0        D    192.168.30.1       GigabitEthernet
0/0/2.30
    192.168.30.1/32   Direct  0     0        D    127.0.0.1         GigabitEthernet
0/0/2.30
    192.168.30.255/32 Direct  0     0        D    127.0.0.1         GigabitEthernet
0/0/2.30
255.255.255.255/32  Direct  0     0        D    127.0.0.1         InLoopBack0
  
```

Q3-2.3. Can PC-10-1 and PC-10-3 ping each other? Please paste the screenshot of the ping result.

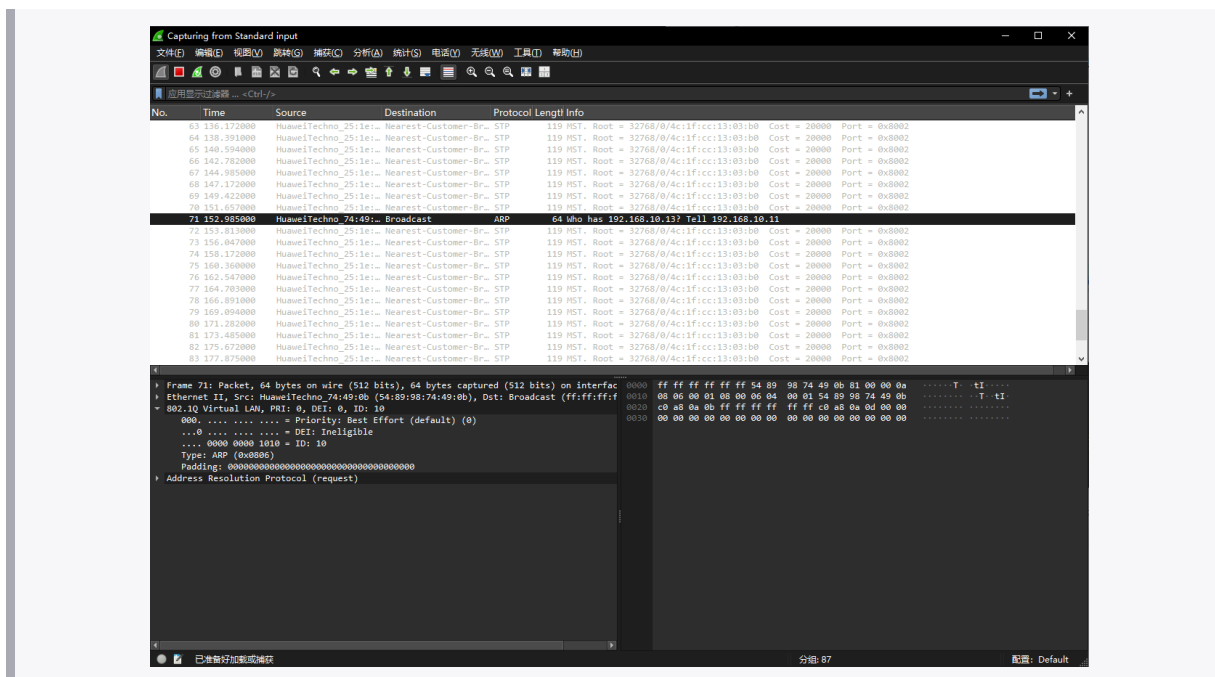
可以互相 ping 通。



Q3-2.4. Paste the screenshot of the result of command “tracert 192.168.30.13” issued from PC-10-1.



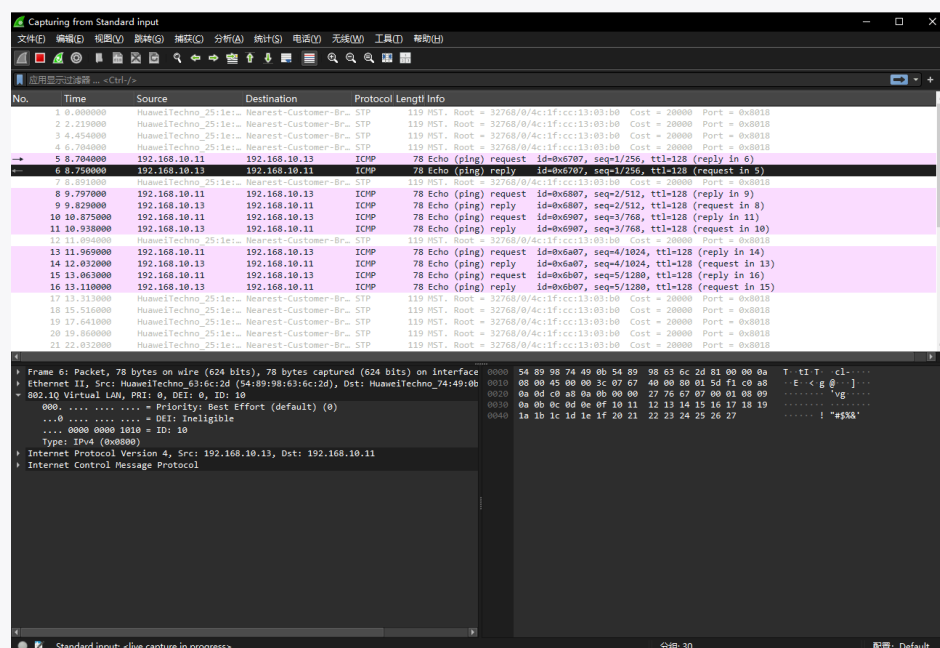
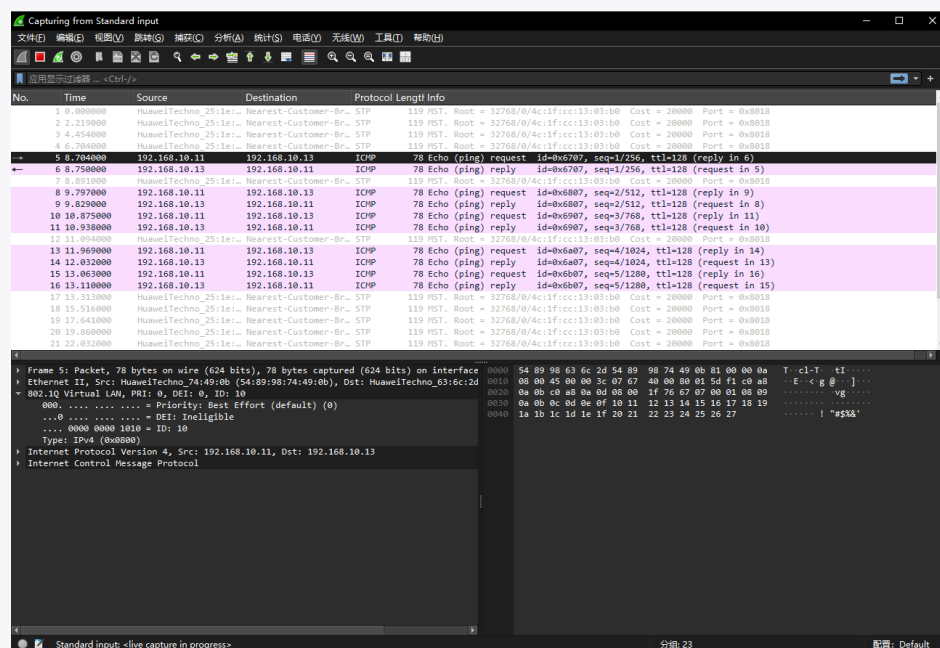
Q3-2.5. Analyze the captured ping traffics between the PC-10-1 and PC-10-3. What are the tagged VLAN IDs of the Ethernet frames leaving and entering the interface GE 0/0/2 on switch LSW3 ? Please paste the screenshot of the all field information of the captured Ethernet frame.



抓取到的 ARP 广播报文携带的 VLAN ID 为 10。由于 PC-10-1 和 PC-10-3 同属于 VLAN 10，它们之间的 Ping 通信在二层交换机（LSW1、LSW3、LSW2）内部根据 MAC 地址表就可以直接完成转发，根本不需要经过路由器。因此，属于单播的 ICMP 数据流不会被上送到连接路由器的 GE 0/0/2 接口。只有像 ARP 这样的广播报文才会被交换机泛洪到该接口。

Q3-2.6. Analyze the captured ping traffics between the PC-10-1 and PC-10-3. What are the tagged VLAN IDs of the Ethernet frames leaving and entering the interface GE 0/0/24 on switch LSW3 ? Please paste the screenshot of the all field information of the captured Ethernet frame.

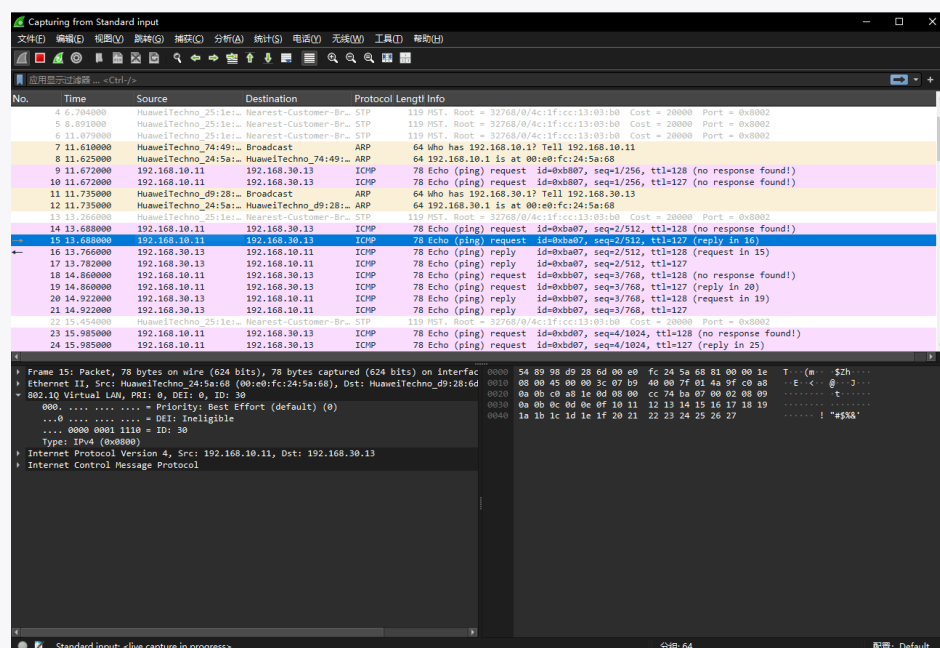
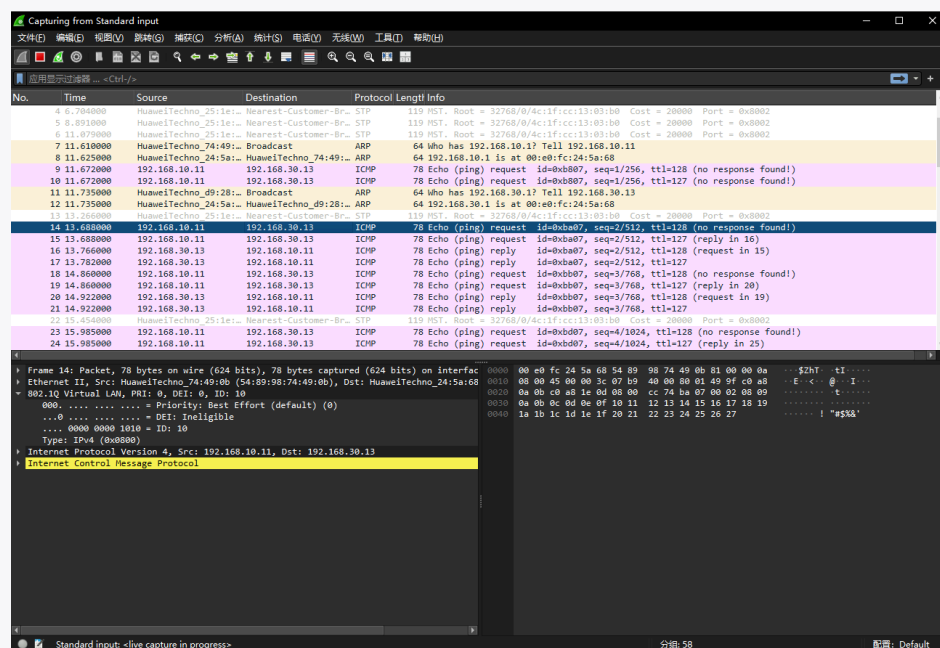
下面分别是进入 LSW3 端口和离开 LSW3 端口的数据帧抓包信息：



捕获到的 Ethernet 帧进入和离开接口 GE 0/0/24 的 VLAN ID 均为 10。由于通信双方处于同一 VLAN 内，数据在跨越交换机的 Trunk 链路上传输时，仅携带其原始的 VLAN 10 标签，不涉及三层路由的解封封装和重封装。

Q3-2.7. Analyze the captured ping traffics between the PC-10-1 and PC-30-3. What are the tagged VLAN IDs of the Ethernet frames leaving and entering the interface GE 0/0/2 on router RTA? Please paste the screenshot of the all field information of the captured Ethernet frame.

下面分别是跨 VLAN 通信时，进入路由器 RTA 端口和离开路由器 RTA 端口的同一个 ICMP 请求报文的抓包信息：



- **进入接口：** 源自 PC-10-1 的报文通过 LSW3 透传到达 RTA 的子接口 0/0/2.10，此时捕获到的 Ethernet 帧带有源网络的标签，VLAN ID 为 10。
- **离开接口：** RTA 接收该报文后，剥离 VLAN 10 的标签，查询路由表发现目的地属于 192.168.30.0/24 网段。于是该报文被转交由子接口 0/0/2.30 发出，并重新打上 VLAN 30 的标签。因此在报文离开该接口时，捕获到的 Ethernet 帧 VLAN ID 变成了 30。